ClawbackVault Engineering Case Study, Second Round

Full Stack AI Engineer | 2 days from receipt | about 5 hours of work | 20-minute review with our team

The problem

Insurance and mortgage brokers lose income when clients quietly leave: the renewal is gone and commission can be clawed back. Clients rarely announce it. The signs show up first in their email: replies get shorter and more formal, a spouse or a CFO appears on the thread, questions go unanswered, or the conversation stops. ClawbackVault connects read-only to a broker's mailbox, watches only the client relationships the broker approves, and each morning tells the broker which relationships need a call, and why.

Your task

Design the architecture of the Relationship Signal Engine: the system that turns a broker's Gmail or Microsoft 365 mailbox into a short, honest morning list of at-risk relationships. It must detect four kinds of signal (explicit: "we have a cheaper quote"; implicit: tone and formality change, new people copied in; drift: contact cadence departing from that relationship's normal; silence: the client has gone quiet), plus unanswered client questions, and give a plain-language reason for every flag.

Non-negotiable constraints

Watchlist gate	No message body is fetched unless it belongs to a relationship the broker approved.
Mask first, fail closed	Names, numbers and identifiers are masked before any AI model sees text. If masking cannot be confirmed complete, nothing is classified.
No content stored	Bodies and subject lines are never stored or logged. Derived facts (signals, dates, reasons) may be.
Tenant isolation	One firm can never read another firm's data, even if application code has a bug.
Email is untrusted	Text inside an email ("ignore your instructions...") must never change what the system does.
Honest output	No number or claim the data does not support. A partial scan says it is partial.

Assumptions to design against

- 300 firms and 1,500 broker seats at launch, 5,000 seats in 18 months; about 120 emails a day per seat, roughly 20% from approved clients.
- Provider limits: about 250 header reads a minute per mailbox; access can be revoked by the broker at any time.
- Digest ready by 07:00 local time. New brokers get a one-time scan of the last 90 days of headers.
- AI model spend under 3 US dollars per seat per month. A team of two to three engineers builds and runs it.

What to submit

- **Architecture document, at most 4 pages plus diagrams:** components and data flow from mailbox to digest; where each constraint is enforced; data model and how tenancy is enforced; the masking pipeline and its failure path; which signals use rules or statistics and which use an AI model, and why; scheduling, rate limiting and recovery when a nightly run fails halfway; your top security threats and

controls; how you would measure whether it works when you have no labelled data; a cost estimate per seat; and what you would build in the first six weeks and deliberately leave out.

- **Optional prototype (recommended):** a small runnable pipeline against the synthetic dataset provided (clawbackvault_case_dataset.json, all fictional) that applies the gate, masks, detects at least one statistical and one language signal, and writes a per-broker digest citing message IDs only, never message text. A stubbed AI model behind a clear interface is fine.

The review session (20 minutes)

A 20-minute session with our team: about 8 minutes for you to walk us through your key decisions and trade-offs, then about 12 minutes of our questions on your design and, if you built one, your prototype.

How we assess

Architecture judgement 30%, privacy and security 25%, AI and evaluation thinking 20%, prototype and code quality 15%, communication 10%. We value clear choices sized for a small team over breadth. AI coding tools are welcome; tell us how you used them and be ready to explain every line and sentence. Questions are welcome in the first 12 hours; after that, state your assumptions and carry on. Use only the synthetic data, never a real mailbox.